$z\dfrac{\partial z}{\partial x} + \left(z^2 - x^2\right)\dfrac{\partial z}{\partial y} = -x.\ \dfrac{dx}{z} = \dfrac{dy}{z^2 - x^2} = \dfrac{dz}{-x}.$

$\dfrac{dx}{z} = \dfrac{dz}{-x}.\ x^2 + z^2 = A.$ Далее скомбинируем $\dfrac{dy}{z^2 - x^2} = \dfrac{z\,dx + x\,dz}{z^2 - x^2}.\ zx - y = B.$

Подставив $\begin{cases} y = x^2 \\ z = 2x \end{cases}$ в $\begin{cases} x^2 + z^2 = A \\ zx - y = B, \end{cases}$ получаем $\begin{cases} 5x^2 = A \\ x^2 = B. \end{cases}\ A = 5B.\ x^2 + y^2 = 5(zx - y).$